

Shortcut Method

New mean = Previous mean $\times$ Constant multiplied to each term $= 18 \times 3 = 54$

2. **Mean of grouped or continuous observations** If $x_1, x_2, x_3, \dots, x_n$ are n observations whose corresponding frequencies are $f_1, f_2, f_3, \dots, f_n$, then

(i) Direct Method

$$(a) \text{ Mean, } \bar{x} = \frac{x_1 f_1 + x_2 f_2 + \dots + x_n f_n}{f_1 + f_2 + \dots + f_n}$$

$$= \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = \frac{\sum f_i x_i}{\sum f_i}$$

$$(b) \text{ Mean, } \bar{x} = A + \frac{\sum_{i=1}^n f_i d_i}{\sum_{i=1}^n f_i} = A + \frac{\sum f_i d_i}{\sum f_i}$$

where, A = Assumed mean and $d_i = x_i - A$.

$$(ii) \text{ Step Deviation Method } (\bar{x}) = A + \left(\frac{\sum f_i u_i}{\sum f_i} \right) \times h$$

where, h = Difference between two consecutive class marks.

EXAMPLE 8. The arithmetic mean of the marks from the following table is

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Number of students	12	18	27	20	17	6

- a. 20 b. 28 c. 2800 d. 100

Sol. b. We have,

Marks	Class mark, (x)	f	fx
0-10	5	12	60
10-20	15	18	270
20-30	25	27	675
30-40	35	20	700
40-50	45	17	765
50-60	55	6	330
Total		$\Sigma f = 100$	$\Sigma fx = 2800$

$$\therefore \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{2800}{100} = 28$$

EXAMPLE 9. Find the mean wage from the data given below :

Wages (in ₹)	800	820	860	900	920	980	1000
Number of workers	7	14	19	25	20	10	5

- a. 890 b. 890.5 c. 891.2 d. 100

Sol. c. Let the assumed mean be $A = 900$.
The given data can be written as under :

Wages (in ₹) (x_i)	Number workers (f_i)	$d = x_i - A$	$f_i d_i$
800	7	-100	-700
820	14	-80	-1120
860	19	-40	-760
900	25	0	0
920	20	20	400
980	10	80	800
1000	5	100	500
Total	$\Sigma f_i = 100$		$\Sigma f_i d_i = -880$

Here, $A = 900$

$$\therefore \text{Mean} = A + \frac{\Sigma f_i d_i}{\Sigma f_i} = 900 + \frac{-880}{100} = 900 - 8.80 = 891.2$$

$$\therefore \text{Mean wage} = ₹891.2$$

EXAMPLE 10. Find the mean of the following data.

x	15	25	35	45	55	65
f	4	28	15	20	17	16

- a. 44.2 b. 42.4 c. 40.6 d. 41.6

Sol. d. Here, the values of class marks x_i are given, so let $A = 35$ and class width, h = difference between two consecutive class marks = 10, then

x_i	f_i	$u_i = \frac{x_i - A}{h}$	$f_i u_i$
15	4	$\frac{15 - 35}{10} = \frac{-20}{10} = -2$	$4 \times (-2) = -8$
25	28	$\frac{25 - 35}{10} = \frac{-10}{10} = -1$	$28 \times (-1) = -28$
35 = (A)	15	$\frac{35 - 35}{10} = \frac{0}{10} = 0$	$15 \times 0 = 0$
45	20	$\frac{45 - 35}{10} = \frac{10}{10} = 1$	$20 \times 1 = 20$
55	17	$\frac{55 - 35}{10} = \frac{20}{10} = 2$	$17 \times 2 = 34$
65	16	$\frac{65 - 35}{10} = \frac{30}{10} = 3$	$16 \times 3 = 48$
Total	$\Sigma f_i = 100$		$\Sigma f_i u_i = 66$

Thus, we have $\Sigma f_i u_i = 66$, $\Sigma f_i = 100$, $h = 10$

$$\therefore \bar{x} = A + h \left(\frac{\Sigma f_i u_i}{\Sigma f_i} \right) = 35 + 10 \times \frac{66}{100}$$

$$= 35 + \frac{660}{100} = 35 + 6.6 \Rightarrow \bar{x} = 41.6$$

Weighted Arithmetic Mean

If corresponding weight of $x_1, x_2, \dots, x_n$ are $w_1, w_2, \dots, w_n$ respectively, then

Weighted arithmetic mean

$$= \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

EXAMPLE 11. The weighted arithmetic mean of the first n natural numbers, the weights being the corresponding numbers, is

- a. $\frac{n+1}{2}$ b. $\frac{n+2}{2}$ c. $\frac{2n+1}{3}$ d. None of these

Sol. c. First n natural numbers are $1, 2, 3, \dots, n$; whose corresponding weights are $1, 2, 3, \dots, n$, respectively.

$$\begin{aligned} \therefore \text{Weight arithmetic mean} &= \frac{1 \times 1 + 2 \times 2 + \dots + n \times n}{1 + 2 + \dots + n} \\ &= \frac{1^2 + 2^2 + \dots + n^2}{1 + 2 + \dots + n} = \frac{\frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)}{2}} = \frac{2n+1}{3} \end{aligned}$$

Combined Arithmetic Mean

If two sets of observations are given, then the combined mean of both the sets can be calculated by the following formula.

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

where, $\bar{x}$ = Mean of sets of observations

$\bar{x}_1$ = Mean of first set of observations

n_1 = Number of observations in first set

$\bar{x}_2$ = Mean of second set of observations

n_2 = Number of observations in second set

EXAMPLE 12. The average salary of male employees in a firm was ₹ 5200 and that of females was ₹ 4200. The mean salary of all the employees was ₹ 5000. The percentage of male and female employees are, respectively.

- a. 80 and 20 b. 20 and 80 c. 60 and 40 d. 52 and 48

Sol. a. Let $x_1 = 5200$, $x_2 = 4200$ and $\bar{x} = 5000$

$$\text{We know that, } \bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\Rightarrow 5000(n_1 + n_2) = 5200n_1 + 4200n_2 \Rightarrow \frac{n_1}{n_2} = \frac{4}{1}$$

$\therefore$ The percentage of male employees in the firm

$$= \frac{4}{4+1} \times 100 = 80$$

and the percentage of female employees in the firm

$$= \frac{1}{4+1} \times 100 = 20$$

Geometric Mean

If a , G and b are in GP, then the geometric mean between a and b is, $G = \sqrt{ab}$.

EXAMPLE 13. What is the geometrical mean of the variate which takes values 210, 201, 202, 20, 12, 10, 2, 1 and 0?

- a. 10 b. 9 c. 8 d. 0

Sol. d. The given variates are 210, 201, 202, 20, 12, 10, 2, 1 and 0.

$$\begin{aligned} \therefore \text{Geometric mean of given variates} \\ = \sqrt[9]{210 \times 201 \times 202 \times 20 \times 10 \times 2 \times 1 \times 0} = \sqrt[9]{0} = 0 \end{aligned}$$

Harmonic Mean

The harmonic mean of n positive numbers $a_1, a_2, \dots, a_n$ is

$$\frac{1}{H} = \frac{1}{n} \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right)$$

where, H denotes the harmonic mean.

Median

After arranging the given data in ascending or descending order of magnitude, the value of the middle most observation is called the median of the data.

1. **Median of an individual series** If number of observations is n . Then, arrange the observations in ascending or descending order.

(i) If n is an odd number, then

$$\text{Median} = \text{Value of the } \left(\frac{n+1}{2} \right) \text{th observation}$$

(ii) If n is an even number, then

$$\text{Value of the } \left(\frac{n}{2} \right) \text{th observation}$$

$$+ \text{Value of } \left(\frac{n}{2} + 1 \right) \text{th observation}$$

$$\text{Median} = \frac{\quad}{2}$$

2. **Median of a discrete frequency series** First arrange the data in ascending or descending order and find cumulative frequency. Now, find $\frac{N}{2}$, where $N = \Sigma f_i$.

See the cumulative frequency just greater than $\frac{N}{2}$.

The corresponding value of x is median.

3. **Median of a continuous series** In this case, the class corresponding to the cumulative frequency just greater than $N/2$ is called the median class and the value of median is obtained by the following formula.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - c}{f} \right) \times b$$

where, l = Lower limit of median class

f = Frequency of median class
 h = Size of median class
 c = Cumulative frequency of class before median class

EXAMPLE 14. From the data given, the median of the average deposit balance of saving for the branch during March 1982 is

Average deposit balance (in ₹)	Number of deposit
0 – 100	26
100-200	68
200-300	145
300-400	242
400-500	188
500-600	65
600-700	16

a. 356 b. 300 c. 56.2 d. 356.2

Sol. d.

Average deposit balance (in ₹)	f	Cumulative frequency (cf)
Less than 100	26	26
100-200	68	94
200-300	145	239
300-400	242	481
400-500	188	669
500-600	65	734
600-700	16	750

$$\frac{N}{2} = \frac{750}{2} = 375$$

The frequency just greater than 375 is 481.

∴ Median class is 300-400.

$$\begin{aligned} \text{Median} &= l + \frac{\left(\frac{N}{2} - c\right)}{f} \times h = 300 + \frac{375 - 239}{242} \times 100 \\ &= 300 + 56.2 = 356.2 \end{aligned}$$

Mode

The data value which occurs maximum number of times in a data distribution is called mode.

- Mode of individual series** The value which is repeated maximum number of times is called mode of the series.

EXAMPLE 15. Find the mode for the following series 2.5, 2.3, 2.2, 2.2, 2.2, 2.4, 2.7, 2.7, 2.5, 2.3, 2.2, 2.6 and 2.2.

a. 2.2 b. 2.3
c. 2.7 d. 2.6

Sol. a. Arranging the data in the form of a frequency table, we have

Value	Frequency
2.2	4
2.3	2
2.4	1
2.5	2
2.6	1
2.7	2

We see that, the value 2.2 has the maximum frequency 4. So, the mode for the given series is 2.2.

- Mode of a discrete frequency series** In this case, mode is the value of the variate corresponding to the maximum frequency.

EXAMPLE 16. Compute the modal value for the following frequency distribution.

x	95	105	115	125	135	145	155	165	175
f	4	2	18	22	21	19	10	3	2

a. 115 b. 125 c. 22 d. 120

Sol. b. From the given table, it is clear that 125 has the highest frequency i.e. 22. Hence, modal value of the given frequency distribution is 125.

- Mode of a continuous series** The class which has maximum frequency is called modal class or group. The mode is given by the formula,

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

where, l = Lower limit of modal class

h = Size of class interval

f_1 = Frequency of modal class

f_0 = Frequency of the class preceding the modal class

f_2 = Frequency of the class succeeding the modal class

EXAMPLE 17. The mode of the following distribution is

Class interval	Frequency
0-10	5
10-20	8
20-30	7
30-40	12
40-50	28
50-60	20
60-70	10
70-80	10

a. 46 b. 6.66
c. 46.67 d. None of these

Sol. c. Here, maximum frequency is 28. Thus, the class 40-50 is the modal class. Here, $f_1 = 28, f_0 = 12, f_2 = 20, l = 40$ and $h = 10$

$$\begin{aligned}\therefore \text{Mode} &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h = 40 + \frac{10(28 - 12)}{(2 \times 28 - 12 - 20)} \\ &= 40 + \frac{10 \times 16}{24} = 40 + 6.666 = 46.67 \text{ (approx)}\end{aligned}$$

Relation between Mean, Median and Mode

The empirical relationship between the three measures of central tendency is given by

$$\text{Mode} = 3 (\text{Median}) - 2 (\text{Mean})$$

In case, mean = median = mode, then distribution is said to be symmetric distribution.

EXAMPLE 18. If in a frequency distribution, the mean and median are 20 and 21 respectively, then its mode is approximate by

- a. 24 b. 23 c. 25 d. None of these

Sol. b. Here, mean = 20 and median = 21

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean} = 63 - 40 = 23$$

Some Useful Formulae

If $x_1, x_2, \dots, x_n$ are n observations with their mean M , then deviation d_i is given by $d_i = |x_i - M|$.

- Mean deviation for individual series is given by $\frac{\sum |d_i|}{n}$.
- Mean deviation for discrete series is given by $\frac{\sum f_i |d_i|}{\sum f_i}$.
- Standard deviation (σ) for ungrouped data is given by $\sigma = \sqrt{\frac{\sum |d_i|^2}{n}}$.
- Standard variation for grouped data is given by $\sigma = \sqrt{\frac{\sum f_i |d_i|^2}{n}}$.
- Variance = σ^2
- Coefficient of variation = $\frac{\text{Standard deviation}}{\text{Mean}} \times 100$
- Coefficient of mean deviation = $\frac{\text{Mean deviation}}{\text{Mean}} \times 100$

EXAMPLE 19. The mean deviation of the data, 3, 5, 6, 7, 8, 10, 11, 14 is

- a. 4 b. 3.25 c. 2.75 d. 2.4

Sol. c. Mean = $\frac{3+5+6+7+8+10+11+14}{8} = \frac{64}{8} = 8$

$$\begin{aligned}\therefore \sum \delta &= |3-8| + |5-8| + |6-8| + |7-8| + |8-8| \\ &\quad + |10-8| + |11-8| + |14-8| = 22\end{aligned}$$

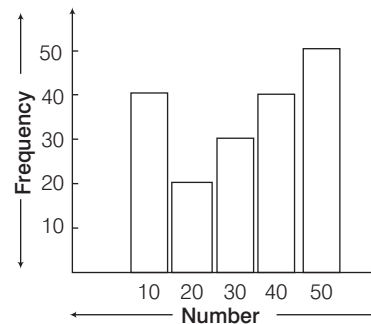
$$\therefore \text{Mean deviation} = \frac{\sum \delta}{n} = \frac{22}{8} = 2.75$$

Graphical Representation of Data

There are many ways of picturing a frequency distribution of continuous type which are discussed below

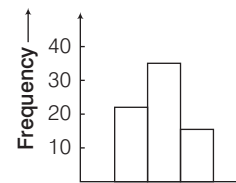
1. Bar Graph

In a bar graph, bars of uniform width are drawn with various heights. The heights of a bar represents the frequency of observation.



2. Histogram

A histogram is the graphical representation of a frequency distribution in the form of the rectangles with class intervals as bases, and the corresponding frequencies as height. There being no gap between any two consecutive rectangles.



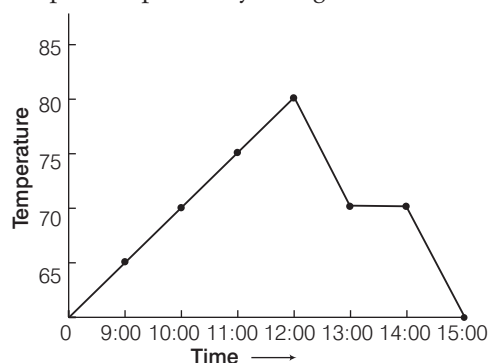
POINTS TO BE REMEMBERED

- A histogram consists of a set of adjacent rectangles, whose bases are equal to class sizes and height are equal to class frequencies.
- The total area of the histogram = Sum of areas of all rectangles. If the class intervals are of same size (width), then the area of histogram = NK , where K = Size of a class and N = Sum of frequencies of all classes.
- If the frequency distribution is discontinuous (inclusive), change it to continuous (exclusive) and then construct a histogram.

3. Frequency polygon

A frequency polygon can be drawn by joining the mid-points of the respective tops of the rectangles of a histogram in case of equal class intervals.

A frequency polygon for a grouped data can also be drawn independently by plotting the mid-points of the all classes along X -axis and frequencies along Y -axis and joining the plotted points by straight line.



4. Ogive (Cumulative Frequency Curve)

When we plot the upper or lower class limits along X -axis and cumulative frequencies along Y -axis, And on joining them we get a curve called an ogive.

We have two types of ogive curves

(a) less than ogive (i.e. the rising curve)

It is the graph between upper limits and cumulative frequencies of a distribution.

(b) more than ogive (i.e. a falling curve)

It is the graph between lower limits and cumulative frequencies of a distribution.

- Note**
- If we have both ogive (i.e. less than type and more than type), then these two ogives intersect each other at a point.
 - From this point, draw a perpendicular on X -axis, the point at which it cuts the X -axis gives the median. i.e. the x -coordinate of intersection point gives the median.

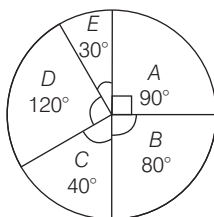
Pie-Diagram (or Pie chart)

A type of graph in which a circle is divided into sectors such that each sector represents a proportion of the whole.

Thus, in a pie chart

- Data is represented by sectors of a circle.
- Each part of data makes a certain central angle.
- Sum of all the angles of sector is 360° .

- Central angle = $\left(\frac{\text{Frequency} \times 360}{\text{Total frequency}} \right)^\circ$

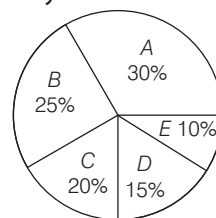


EXAMPLE 20. In statistics, a suitable graph for representing the partitioning of total into subpart is

- a. an ogive
- b. a pictograph
- c. a histogram
- d. a pie chart

Sol. d. In statistics, a suitable graph for representing the partitioning of total into sub parts is a pie chart.

EXAMPLE 21. The following diagram show the expenditure of a family on various items A, B, C, D and E .



Study the diagram carefully and answer the following questions.

(i) The angle of pie diagram showing the expenditure incurred on item A is

- a. 30°
- b. 35°
- c. 108°
- d. None of these

(ii) Which two expenditures together will form an angle of 90° at the centre of pie diagram?

- a. B and C
- b. C and A
- c. D and E
- d. None of these

(iii) If the income of the family is ₹ 3000 per month, then expenditure of item C will be

- a. ₹ 400
- b. ₹ 500
- c. ₹ 600
- d. ₹ 800

Sol. (i) c. Angle for $100\% = 360^\circ$

$$\therefore \text{Angle for expenditure of } A = 30\% \text{ of } 360^\circ \\ = \frac{30}{100} \times 360 = 108^\circ$$

(ii) c. $\because 100\% = 360^\circ$

$$\therefore \text{An angle of } 90^\circ \text{ at the centre of pie diagram in percentage is } = \frac{90 \times 100}{360} = 25\%$$

Expenditure of items D and E makes upto $(15 + 10) = 25\%$.

So, expenditure of D and E together will form an angle of 90° at the centre of pie diagram.

(iii) c. Expenditure on item C

$$= 20\% \text{ of } ₹ 3000 = \frac{20}{100} \times 3000 = ₹ 600$$